

L2 - 4.1. Monitoring Active Directory

Task 1 - Introduction

AD is the identity backbone of most enterprise networks, making it a primary target for attackers. AD generates thousands of events per hour, including authentication requests, group changes, service tickets, and failed logins. Most of this activity is completely normal.

Task 2 - AD Traffic and Logging

Protocol Summary

Protocol	Ports	What It Does	Normal Usage
Kerberos	88	Default authentication in AD	User logins, service access, ticket requests
LDAP	389, 636, 3268, 3269	Directory queries and modifications	User lookups, group membership checks, and address book queries
SMB	445 (139 for legacy NetBIOS sessions)	File sharing, remote administration	Accessing shared folders, printers, and administrative tools
RDP	3389	Interactive remote desktop	Help desk support, server administration
Name Resolution (Legacy)	137, 138, 5355	NetBIOS (TCP/UDP - 137/138) and LLMNR (5355)	Fallback when DNS fails, for older applications

Task 3 - Authentication Events

In this room, we focus on domain authentication using two protocols: modern Kerberos and legacy NTLM.

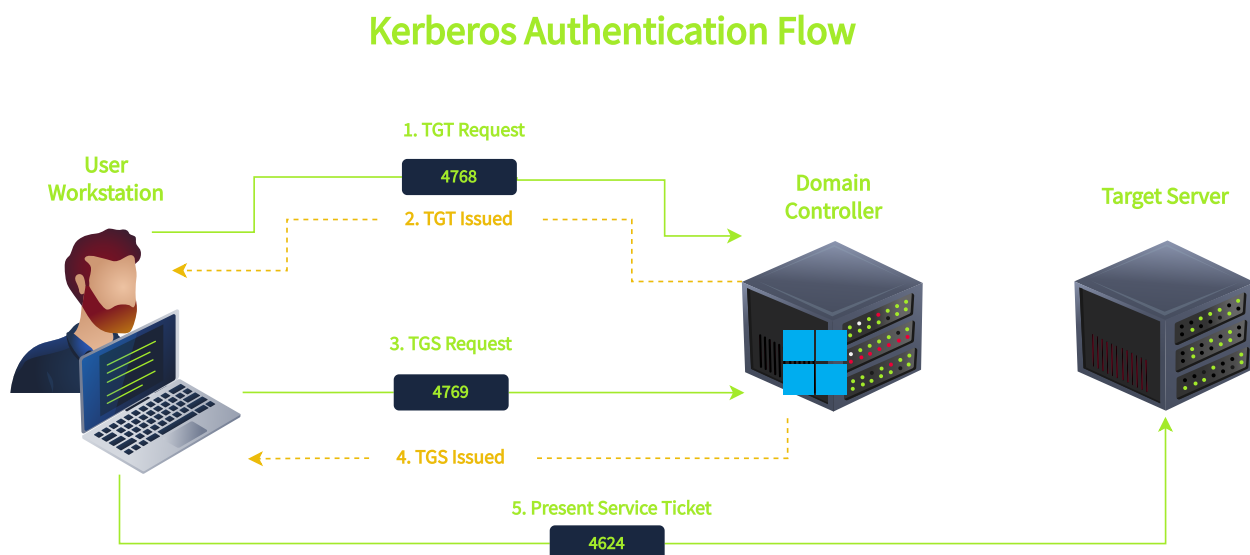
Domain Users vs Local Users

- **Domain users** authenticate against the Domain Controller. Their credentials are stored in the AD database (`NTDS.dit`). When they log in, the DC is involved, so events appear on the DC.
- **Local users** authenticate against the local **Security Account Manager** (`SAM`) database on each machine. The DC isn't involved, so events only appear on that specific machine.

User Type	Credentials Stored	Where Authentication Events Appear
Domain user	NTDS.dit (on DC)	Domain Controller
Local user	SAM (on local machine)	Local machine only

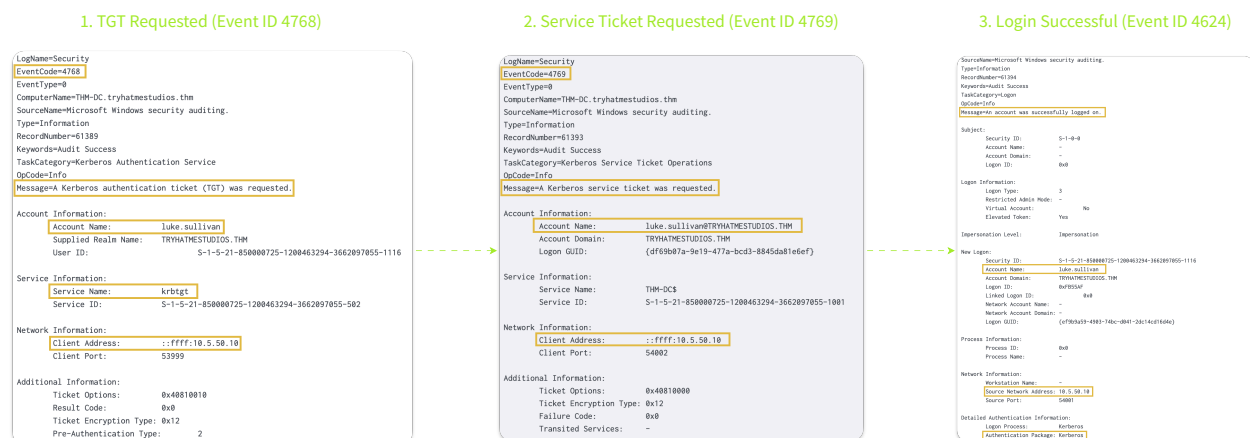
Domain Controller logs provide a centralized, domain-wide view of user activity across multiple workstations, whereas local logins are confined strictly to individual workstation security logs.

Kerberos Authentication: Kerberos is a ticket-based authentication protocol used by Active Directory to verify user identity without sending passwords over the network. When a user authenticates, they first request a Ticket Granting Ticket (TGT) from the Domain Controller. This TGT is then used to request service tickets (TGS) for specific resources. Each step generates a particular event:



Step	What Happens	Event ID	Where Logged
1	User requests a TGT	4768	Domain Controller
2	User requests a TGS	4769	Domain Controller
3	User creates a session on the target	4624	Target server

This is how it appears in logs:



What if the user enters the wrong password or the authentication fails for any other reason? The pre-authentication step fails, generating Event 4771 on the Domain Controller. We'll see 4771 events regularly in environments where users sometimes forget or mistype their passwords.

Encryption Types in 4768/4769: Tickets usually are encrypted, and AD uses two encryption types for this:

- **RC4** encryption appears in environments with older systems or applications that don't support AES.
- And **AES-256** encryption, which modern systems use.

Understanding what should usually appear in our environment helps us determine which encryption types to expect and where they come from.

Value	Algorithm	When You See It
0x12	AES-256	Modern systems, Windows 2008+ domain functional level
0x17	RC4-HMAC	Legacy systems, older applications, cross-forest trusts

TGT requests in Splunk

```
index=* EventCode=4768
| table _time, Account_Name, Client_Address, Ticket_Encryption_Type
```

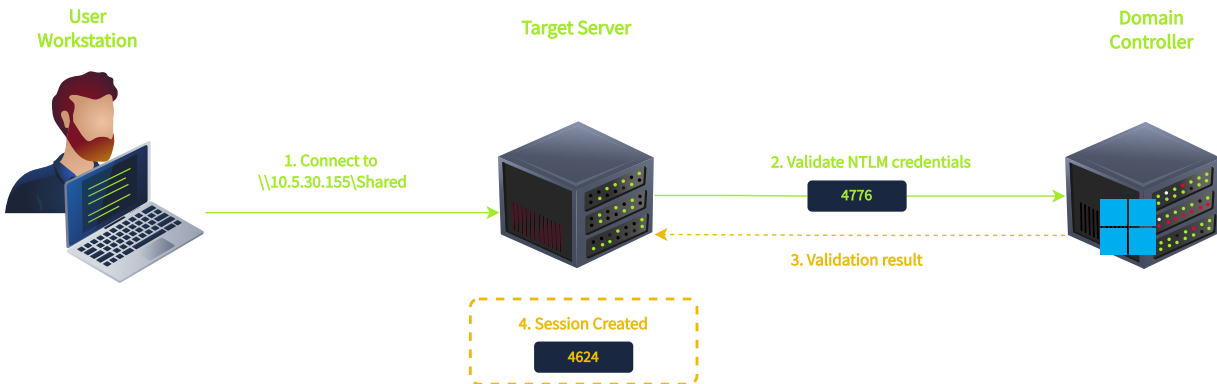
This shows us:

- Who tried to authenticate (`Account_Name`),
- When (`_time`),
- From which machine (`Client_Address`),
- And the encryption type for the ticket requested (`Ticket_Encryption_Type`).

NTLM Authentication: NTLM is a legacy authentication protocol used when Kerberos is unavailable. This happens when accessing resources by IP address, when the target system cannot be found in DNS, or during authentication to non-domain systems.

Step	What Happens	Event ID	Where Logged
1	The target server asks the DC to validate credentials	4776	Domain Controller
2	Session created on target	4624	Target server

NTLM Authentication Flow



Event 4776 appears on the Domain Controller when NTLM credentials are validated. Common scenarios include:

- Accessing file shares by IP address (\\10.0.1.50\Shared)
- Legacy applications that don't support Kerberos
- Authentication across untrusted domains

To view NTLM authentication attempts on the DC in Splunk:

```
index=* EventCode=4776
| table _time, Logon_Account, Source_Workstation
```

This shows us:

- Which account was authenticated (Logon_Account),
- From which machine (Source_Workstation).

Meanwhile, on the target host, it appears as follows:

```
index=* EventCode=4624 Account_Name=michelle.smith Authentication_Package=NTLM
| table _time host user Workstation_Name Source_Network_Address Authentication_Package
```

NTLM authentication is less common than Kerberos in modern environments. If we see high volumes of Event 4776, this may indicate misconfigured systems,

legacy applications, or services accessing resources by IP instead of hostname.

Answer the questions below

| Which file stores domain user credentials on the domain controller?

NTDS.dit

| A local user authenticates to a workstation. Will this generate any events on the Domain Controller?

Nay

| What Event ID is generated when a user requests a TGT?

4768

| In win index in Splunk, how many unique accounts requested TGTs in the dataset across all time?

14

```
index=win EventCode=4768
| dedup Account_Name
| table _time, Account_Name, Client_Address, Ticket_Encryption_Type
```

Task 4 - Accounts, Groups and Resource Access Events

Authentication events tell us who connected.

Account Lifecycle Events: Every user account in Active Directory goes through a lifecycle:

- Account creation,
- Password resets,
- Occasional lockouts,
- And eventually, deactivation when someone leaves.

These routine administrative actions generate specific Event IDs that create a baseline of normal activity.

Event ID	What Happened
4720	Account created
4722	Account enabled
4724	Password reset attempted
4725	Account disabled
4740	Account locked out

In a typical environment, account lifecycle events follow predictable patterns:

- When HR onboards a new employee, IT creates their account (4720) and enables it (4722).
- When the employee forgets their password, the help desk resets it (4724).
- When they eventually leave the company, IT deactivates the account (4725).

To see what accounts were created in our environment, when, and by whom, we can use this Splunk query:

```
index=* EventCode=4720
| table _time, SAM_Account_Name, Subject_Account_Name
```

The `SAM_Account_Name` field shows the account that was created, while `Subject_Account_Name` shows which admin account created it.

Group Membership Events: Adding someone to a privileged group is one of the fastest ways an attacker can escalate from an initial foothold to domain-wide control and establish persistence. Fortunately, Windows logs every security group membership change with dedicated Event IDs.

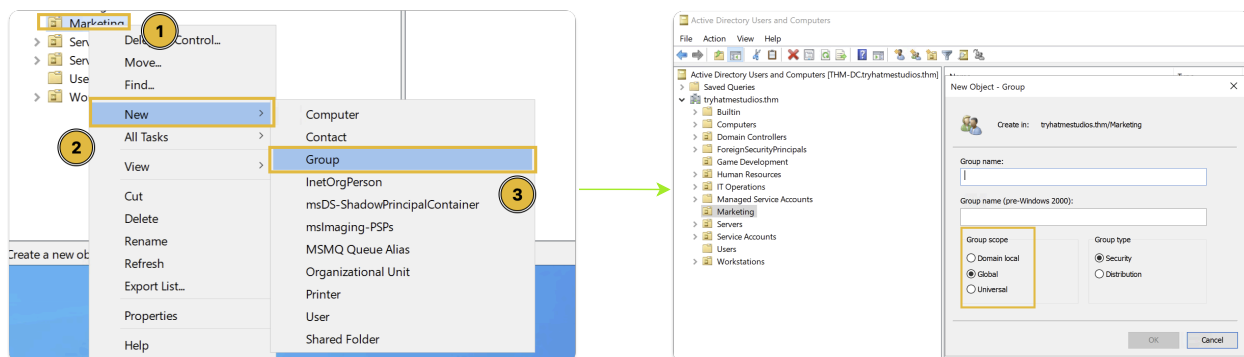
Event ID	What Happened	Group Scope
4728	Member added to the global security group	Domain-wide
4732	Member added to local security group	Machine-level (domain local on DCs)

Event ID	What Happened	Group Scope
4756	Member added to universal security group	Entire forest

Active Directory uses different group scopes (global, domain local, and universal) to organize users and control access. The naming can be confusing (e.g., global groups are actually scoped to a single domain, not globally). For monitoring purposes, what matters is that **all three Event IDs indicate someone was added to a security group**. Watch all of them, especially for privileged groups like Domain Admins, Enterprise Admins, and local Administrators.

LogName=Security EventCode=4732 EventTypeName: ComputerName=THM-DC. tryhatstudios.thm SourceName=Microsoft Windows security auditing. Type=Information RecordNumber=76217 Keywords=Audit Success TaskCategory=Security Group Management OpCode=Info Message= member was added to a security-enabled local group. Subject: Security ID: S-1-5-21-850880725-120463294-3662897855-1128 Account Name: adn-luke.sullivan Account Domain: tryhatstudios Logon ID: 0x162489 Member: Security ID: S-1-5-21-850880725-120463294-3662897855-1115 Account Name: CN=Liam Patel,OU=Game Development,DC=tryhatstudios,DC=thm Group: Security ID: S-1-5-32-555 Group Name: Remote Desktop Users Group Domain: BuiltIn Additional Information: Privileges: -	LogName=Security EventCode=4728 EventTypeName: ComputerName=THM-DC. tryhatstudios.thm SourceName=Microsoft Windows security auditing. Type=Information RecordNumber=64397 Keywords=Audit Success TaskCategory=Security Group Management OpCode=Info Message= member was added to a security-enabled global group. Subject: Security ID: S-1-5-21-850880725-120463294-3662897855-1128 Account Name: adn-luke.sullivan Account Domain: tryhatstudios Logon ID: 0x162489 Member: Security ID: S-1-5-21-850880725-120463294-3662897855-1116 Account Name: CN=Luke Sullivan,OU=IT Operations,DC=tryhatstudios,DC=thm Group: Security ID: S-1-5-21-850880725-120463294-3662897855-2685 Group Name: IT_Team Group Domain: tryhatstudios Additional Information: Privileges: -	LogName=Security EventCode=4756 EventTypeName: ComputerName=THM-DC. tryhatstudios.thm SourceName=Microsoft Windows security auditing. Type=Information RecordNumber=64482 Keywords=Audit Success TaskCategory=Security Group Management OpCode=Info Message= member was added to a security-enabled universal group. Subject: Security ID: S-1-5-21-850880725-120463294-3662897855-1128 Account Name: adn-luke.sullivan Account Domain: tryhatstudios Logon ID: 0x162489 Member: Security ID: S-1-5-21-850880725-120463294-3662897855-1114 Account Name: CN=Jessica Martinez,OU=Human Resources,DC=tryhatstudios,DC=thm Group: Security ID: S-1-5-21-850880725-120463294-3662897855-2686 Group Name: All_Staff Group Domain: tryhatstudios Additional Information: Privileges: -
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On domain controllers, we can see these groups when right-clicking on a specific OU → New → Group:



To see group membership changes for any of these groups, we can run the following Splunk query:

```
index=* (EventCode=4728 OR EventCode=4732 OR EventCode=4756)
| table _time, Member_Account_Name, Group_Name, Subject_Account_Name
```

nt_Name

This shows us:

- Who was added (`Member_Account_Name`),
- Which group they were added to (`Group_Name`),
- And who made the change (`Subject_Account_Name`).

Directory Service Events: Event 5136 logs attribute-level modifications to Active Directory objects. Unlike Event 4728, which only indicates that someone was added to a group, Event 5136 shows the specific LDAP attribute that changed and its new value.

These are the attributes we'll see most often:

- `userAccountControl` - Account status changes (disabled, password never expires, etc.)
- `servicePrincipalName` - SPN modifications
- `scriptPath` - Logon script path (scripts that run when users log in)
- `member` - Group membership modifications at the attribute level
- `displayName`, `description`, `title` - User information fields

For example, when an IT administrator configures a logon script for a user, Event 5136 captures the exact path:

In this case, we can see that a logon script was configured for a user who made the change (`Subject_Account_Name: adm-luke.sullivan`), and the script path. Monitoring these changes helps verify that configuration modifications are made by authorized administrators and align with our change management process.

Tracking GPO Modifications: One important use case for Event 5136 is monitoring changes to Group Policy Objects. GPOs allow administrators to manage configuration across the domain centrally. A single GPO can deploy software, configure security settings, or change audit policies on thousands of machines at once.

To view GPO modifications in Splunk:

```
index=* EventCode=5136 Class="groupPolicyContainer"
| table _time, Subject_Account_Name, DN, LDAP_Display_Name, Value
| sort - _time
```



Note: This query won't return results on the attached VM since no GPO modifications occurred during the captured time window. The screenshot below shows what the output looks like in an environment where GPOs were modified.

_time	Subject_Account_Name	DN	LDAP_Display_Name	Value
2026-02-03 05:44:57	adm-luke.sullivan	CN={AF2E0408-0AB2-4BA8-A4EF-625D773F812F},CN=Policies,CN=System,DC=tryhatmestudios,DC=thm	flags	0
2026-02-03 05:44:57	adm-luke.sullivan	CN={AF2E0408-0AB2-4BA8-A4EF-625D773F812F},CN=Policies,CN=System,DC=tryhatmestudios,DC=thm	gPCFunctionalityVersion	2
2026-02-03 05:44:57	adm-luke.sullivan	CN={AF2E0408-0AB2-4BA8-A4EF-625D773F812F},CN=Policies,CN=System,DC=tryhatmestudios,DC=thm	versionNumber	0
2026-02-03 05:44:57	adm-luke.sullivan	CN={AF2E0408-0AB2-4BA8-A4EF-625D773F812F},CN=Policies,CN=System,DC=tryhatmestudios,DC=thm	gPCFileSysPath	\\tryhatmestudios.thm\SysVol\tryhatmestudios.thm\Policies\{AF2E0408-0AB2-4BA8-A4EF-625D773F812F}
2026-02-03 05:44:57	adm-luke.sullivan	CN={AF2E0408-0AB2-4BA8-A4EF-625D773F812F},CN=Policies,CN=System,DC=tryhatmestudios,DC=thm	objectClass	1.2.840.113556.1.5.157
2026-02-03 05:44:57	adm-luke.sullivan	CN={AF2E0408-0AB2-4BA8-A4EF-625D773F812F},CN=Policies,CN=System,DC=tryhatmestudios,DC=thm	cn	{AF2E0408-0AB2-4BA8-A4EF-625D773F812F}
2026-02-03 05:44:57	adm-luke.sullivan	cn={AF2E0408-0AB2-4BA8-A4EF-625D773F812F},cn=policies,cn=system,DC=tryhatmestudios,DC=thm	displayName	Corp_Security_Baseline_Workstations
2026-02-03 05:44:57	adm-luke.sullivan	cn={AF2E0408-0AB2-4BA8-A4EF-625D773F812F},cn=policies,cn=system,DC=tryhatmestudios,DC=thm	displayName	New Group Policy Object
2026-02-03 05:44:57	adm-luke.sullivan	CN={AF2E0408-0AB2-4BA8-A4EF-625D773F812F},CN=Policies,CN=System,DC=tryhatmestudios,DC=thm	displayName	New Group Policy Object

This query returns all GPO changes, showing us:

- Who modified (`Subject_Account_Name`),
- Which GPO was affected (`DN`),
- And what changed (`LDAP_Display_Name` and `Value`).
 - `gPCFileSysPath` , which shows the `SYSDVOL` for the group policy being configured.
 - `versionNumber` attribute (indicates modification count)

This visibility tells us when GPO objects are created, renamed, or modified.

The `displayName` and `DN` fields help us identify which GPO was touched, and `Subject_Account_Name` shows who made the change.



Note: While Event 5136 captures changes to GPO metadata stored in Active Directory (such as the GPO's name, version number, or `SYSVOL` path), it doesn't capture changes to the policy settings configured within the GPO. For example, if an administrator changes a password policy from 10 characters to 14 characters, Event 5136 will show the GPO's `versionNumber` incrementing, but not what specific setting changed. The actual policy configurations are stored in `SYSVOL` files, which require separate monitoring.

Logon Events: Every time someone authenticates to a system, Windows records whether the authentication succeeded or failed. These events capture interactive logins at a workstation, network connections to file shares, RDP sessions, and service accounts starting background processes.

Understanding logon events helps us answer fundamental questions during investigations:

- Who accessed this system?
- When?
- From where?
- And did they succeed?

Event ID	What Happened
4624	Successful logon
4625	Failed logon

Understanding Logon Types: The `LogonType` field tells us what kind of activity generated the logon event. For example:

- A Type 2 means someone sat down at a keyboard.
- A Type 3 means they accessed a file share remotely.

Knowing the difference helps us interpret whether the activity makes sense for that account and context.

Type	Meaning	Example
2	Interactive	User at keyboard, physical console

Type	Meaning	Example
3	Network	File share access, WMI queries, remote administration
4	Batch	Scheduled tasks running under a user account
5	Service	Windows services starting under a service account
7	Unlock	User unlocking a previously locked workstation
10	RemoteInteractive	RDP session

To see the distribution of logon types in our environment, we run:

```
index=* EventCode=4624
| stats count by Logon_Type
| sort -count
```

Logon_Type	count
3	1311
5	243
7	46
2	12
10	5
11	1

The results show us which logon types are most common.



Tip: In most environments, Type 3 (network) will dominate because file share access and remote administration generate large volumes. Type 2 and Type 10 represent actual user sessions and should be of lower volume.

Answer the questions below

| In Splunk, what field contains the group name?

Group_Name

In Splunk, what is the MOST common logon type in the dataset?

3

Task 5 - Understanding Baseline Activity

Volume is Normal: Looking at AD logs might seem overwhelming. A single user accessing a file share generates multiple events. Multiply that by hundreds or thousands of users, and we're looking at massive event counts. Some rough numbers for a 500-user environment:

Event	Typical Daily Volume
4769 (TGS requests)	50,000 to 100,000+
4768 (TGT requests)	5,000 to 10,000+
4624 (logons)	Varies widely by server role

This volume isn't a problem because it's completely normal. Our job isn't to review every event, but to find the anomalies hiding in the noise.

Computer Accounts (\$): Computer accounts in AD end with a `$` character (e.g., `THM-DC$`, `THM-IT-DESK$`). These accounts authenticate constantly for machine-to-machine communication, system updates, and automated processes. Computer accounts can generate 70-80% of all Kerberos traffic in busy environments. When investigating user activity, we typically want to filter out non-human activity to focus on human behavior.

To see the difference, we can compare the total count of authentication events for computer accounts vs user accounts using the following Splunk query:

```
index=* EventCode IN (4624, 4768, 4769)
| eval AccountType=if(like(Account_Name, "%$"), "Computer Account", "User Account")
| stats count by AccountType, EventCode
| sort AccountType, -count
```

We can see that computer accounts have 1500+ successful login events, while user accounts have 350+ login events. This shows that filtering out these computer accounts can make a huge difference during investigations.

Common Service Name Patterns: When we look at 4769 events, the `Service_Name` field tells us what resource was accessed:

ServiceName Pattern	What It Represents
krbtgt	TGT renewal requests
cifs/THM-SHR-SRV	File share access
ldap/THM-DC	Directory queries
http/THM-WEB-SRV	Web application access
MSSQLSvc/THM-SQL-SRV	SQL Server access
HOST/THM-IT-DESK	General host services

These patterns represent our baseline of normal activity. When we see a service name that doesn't fit these patterns, that's something worth investigating more closely.

Service Account Patterns: Service accounts follow predictable patterns. Same sources, same destinations, same times. Understanding these patterns helps us recognize what's normal for our environment:

Account Type	Expected Pattern
SQL service accounts	Access database servers from application servers
Backup service accounts	Access file shares from backup servers, typically overnight
Monitoring accounts	Query multiple systems at regular intervals
Admin accounts	Access from designated admin workstations during business hours

Stack Counting Technique: Stack counting (also known as long-tail analysis) is one of the most useful techniques for detecting anomalies. It's simply counting how many times each value appears, sorting the results by frequency, and then focusing your attention on the rare events at the bottom of the list. We demonstrate this technique using Event 4769 because it shows service ticket requests, which reveal what resources users accessed and how frequently. High-volume activity is normal, but rare activity often indicates anomalies worth investigating.

For example, to find suspicious accounts using stack counting:

```
index=* EventCode=4769 NOT Account_Name="*$*"
| stats count by Account_Name
| sort -count
```

As shown in the results above:

- **Top results** show our baseline: normal, expected accounts accessing services regularly
- **Bottom results** show rare events: accounts that only appeared once or twice, which is where anomalies typically hide

If 90% of our service ticket requests come from known, authorized accounts in our environment, that's normal. But if we see an account at the bottom that only requested one or two service tickets, that stands out precisely because it's rare and deserves investigation.

This technique applies to any field in any event type, for example:

- Stack count by `Account_Name` to find unusual user accounts
- Stack count by `Client_Address` to find unusual source IPs
- Stack count by `Service_Name` to find unusual services being accessed
- Stack count by `Ticket_Encryption_Type` to find unusual encryption methods

The rare values at the bottom are where we focus our investigation efforts.

Time-Based Patterns: Timing matters when evaluating whether activity is normal. Business applications are accessed during work hours, backup services run overnight, and batch jobs run within scheduled windows. When we see activity, we should ask ourselves whether the timing makes sense for that account and its purpose.

Understanding these patterns for our environment helps us know what to expect:

Activity	Expected Time
User logins	Business hours (8 AM - 6 PM)
Backup account activity	Overnight (midnight - 4 AM)
Batch job accounts	Scheduled windows

Activity	Expected Time
Admin account usage	During maintenance windows

Understanding these patterns helps us recognize when something deviates from the norm.

Answer the questions below

| What character suffix identifies computer accounts in AD?

\$

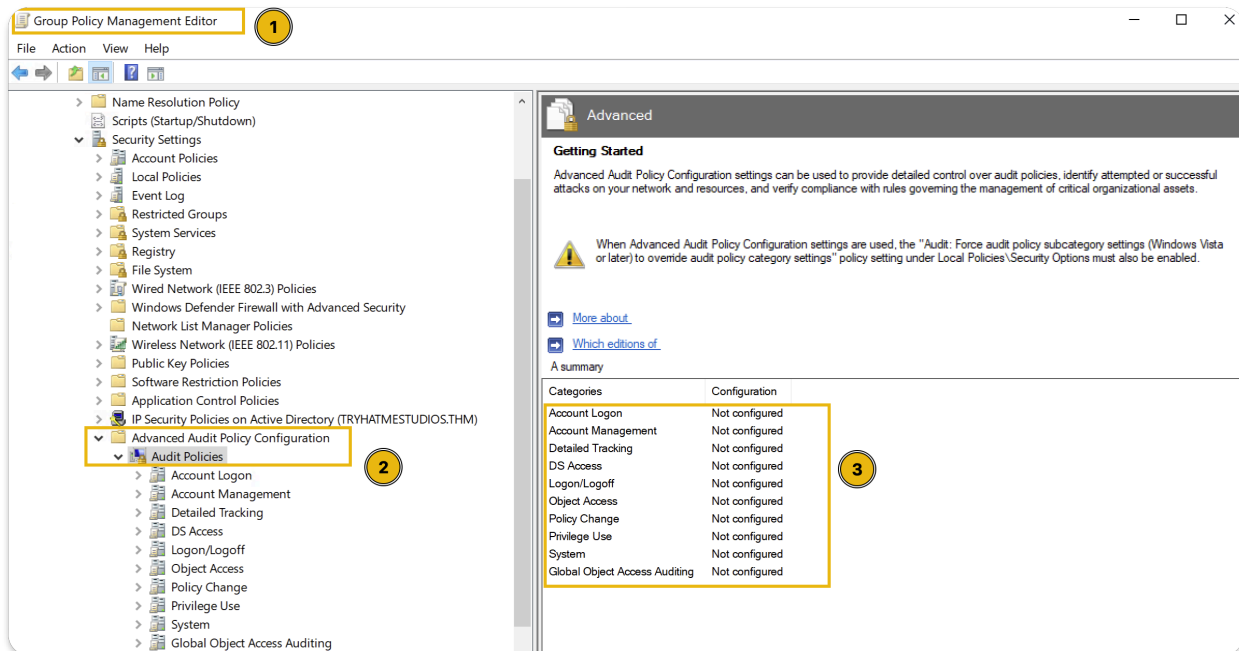
| Using Event ID 4769, what is the MOST frequently requested service?

THM-DC\$

```
index=* EventCode=4769 NOT Account_Name="*$*"
| stats count by Service_Name
| sort -count
```

Task 6 - Audit Policy Configuration

Advanced Audit Policy Configuration: The **Advanced Audit Policy Configuration** is found in Group Policy at: **Computer Configuration** → **Policies** → **Windows Settings** → **Security Settings** → **Advanced Audit Policy Configuration** → **Audit Policies** as shown below:



Here are the minimum settings for full AD visibility:

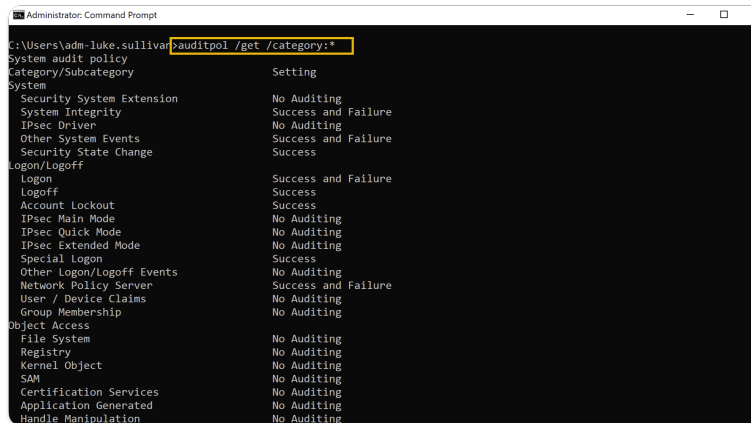
Category	Subcategory	Setting	Events Generated
Account Logon	Credential Validation	Success, Failure	4776 (NTLM)
Account Logon	Kerberos Authentication Service	Success, Failure	4768, 4771
Account Logon	Kerberos Service Ticket Operations	Success, Failure	4769
Account Management	User Account Management	Success, Failure	4720, 4722, 4724, 4725
Account Management	Security Group Management	Success, Failure	4728, 4732, 4756
DS Access	Directory Service Changes	Success, Failure	5136
Logon/Logoff	Logon	Success, Failure	4624, 4625
Object Access	File Share	Success	5140

After we have seen what each of these events captures in the previous tasks, this table should make more sense to us now. Each subcategory controls whether

those specific events get logged.

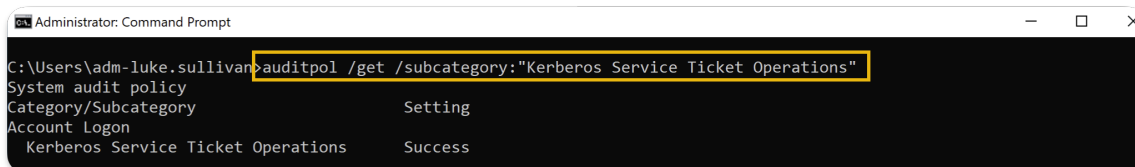
Verifying Your Configuration: In a production environment, we can check current audit settings on a Domain Controller either from **Group Policy Management Editor** as shown above, or by running the following commands:

```
# View all audit policy settings
auditpol /get /category:*
```



Category/Subcategory	Setting
System	
Security System Extension	No Auditing
System Integrity	Success and Failure
IPsec Driver	No Auditing
Other System Events	Success and Failure
Security State Change	Success
Logon/Logoff	
Logon	Success and Failure
Logoff	Success
Account Lockout	Success
IPsec Main Mode	No Auditing
IPsec Quick Mode	No Auditing
IPsec Extended Mode	No Auditing
Special Logon	Success
Other Logon/Logoff Events	No Auditing
Network Policy Server	Success and Failure
User / Device Claims	No Auditing
Group Membership	No Auditing
Object Access	
File System	No Auditing
Registry	No Auditing
Kernel Object	No Auditing
SAM	No Auditing
Certification Services	No Auditing
Application Generated	No Auditing
Handle Manipulation	No Auditing

```
# View specific subcategory
auditpol /get /subcategory:"Kerberos Service Ticket Operations"
```



Category/Subcategory	Setting
Account Logon	
Kerberos Service Ticket Operations	Success

The output shows current audit settings. Each subcategory should show "Success and Failure" for the categories listed above.

Answer the questions below

| What command displays all current audit policy settings on a domain controller?

```
auditpol /get /category:*
```

Task 7 - Scenario: New Employee Onboarding Audit

Scenario: A new employee joined the company's marketing department yesterday. As part of the security team's onboarding verification process, we need to audit their account creation and first-day activity to ensure everything looks normal. Use what you learned in the previous tasks to investigate this user activity in Splunk.

Answer the questions below

| What is the name of the newly created account?

nathan.brooks

```
index=* action=created
```

or

```
index=* EventCode=4720  
| table -time, SAM_Account_Name, Subject_Account_Name
```

```

2/3/26      02/03/2026 01:06:29 PM
9:06:29.000 PM LogName=Security
              EventCode=4720
              EventType=0
              ComputerName=THM-DC.tryhatmestudios.thm
              SourceName=Microsoft Windows security auditing.
              Type=Information
              RecordNumber=71679
              Keywords=Audit Success
              TaskCategory=User Account Management
              OpCode=Info
              Message=A user account was created.

Subject:
  Security ID:      S-1-5-21-850000725-1200463294-3662097055-1120
  Account Name:     adm-luke.sullivan
  Account Domain:   tryhatmestudios
  Logon ID:         0x2352CE2

New Account:
  Security ID:      S-1-5-21-850000725-1200463294-3662097055-2607
  Account Name:     nathan.brooks
  Account Domain:   tryhatmestudios

Attributes:
  SAM Account Name:  nathan.brooks
  Display Name:      Nathan Brooks
  User Principal Name: nathan.brooks@tryhatmestudios.thm
  Home Directory:    -
  Home Drive:        -
  Script Path:       -
  Profile Path:      -
  User Workstations: -
  Password Last Set: <never>
  Account Expires:   <never>
  Primary Group ID:  513
  Allowed To Delegate To: -
  Old UAC Value:     0x0
  New UAC Value:     0x15
  User Account Control:
    Account Disabled
    'Password Not Required' - Enabled
    'Normal Account' - Enabled
  User Parameters:  -
  SID History:      -

```

What group was this user added to?

adm-luke.sullivan

Time	Event																				
2/3/26 9:12:34.000 PM	02/03/2026 01:12:34 PM LogName=Security EventCode=4727 EventType=0 ComputerName=THM-DC.tryhatmestudios.thm SourceName=Microsoft Windows security auditing. Type=Information RecordNumber=71743 Keywords=Audit Success TaskCategory=Security Group Management OpCode=Info Message=A security-enabled global group was created. Subject: <table> <tr> <td>Security ID:</td><td>S-1-5-21-850000725-1200463294-3662097055-1120</td></tr> <tr> <td>Account Name:</td><td>adm-luke.sullivan</td></tr> <tr> <td>Account Domain:</td><td>tryhatmestudios</td></tr> <tr> <td>Logon ID:</td><td>0x2352CE2</td></tr> </table> New Group: <table> <tr> <td>Security ID:</td><td>S-1-5-21-850000725-1200463294-3662097055-2608</td></tr> <tr> <td>Group Name:</td><td>Marketing</td></tr> <tr> <td>Group Domain:</td><td>tryhatmestudios</td></tr> </table> Attributes: <table> <tr> <td>SAM Account Name:</td><td>Marketing</td></tr> <tr> <td>SID History:</td><td>-</td></tr> </table> Additional Information: <table> <tr> <td>Privileges:</td><td>-</td></tr> </table> Collapse host = THM-DC source = WinEventLog:Security sourcetype = WinEventLog	Security ID:	S-1-5-21-850000725-1200463294-3662097055-1120	Account Name:	adm-luke.sullivan	Account Domain:	tryhatmestudios	Logon ID:	0x2352CE2	Security ID:	S-1-5-21-850000725-1200463294-3662097055-2608	Group Name:	Marketing	Group Domain:	tryhatmestudios	SAM Account Name:	Marketing	SID History:	-	Privileges:	-
Security ID:	S-1-5-21-850000725-1200463294-3662097055-1120																				
Account Name:	adm-luke.sullivan																				
Account Domain:	tryhatmestudios																				
Logon ID:	0x2352CE2																				
Security ID:	S-1-5-21-850000725-1200463294-3662097055-2608																				
Group Name:	Marketing																				
Group Domain:	tryhatmestudios																				
SAM Account Name:	Marketing																				
SID History:	-																				
Privileges:	-																				

What group was this user added to?

Marketing

```
index=* (EventCode=4728 OR EventCode=4732 OR EventCode=4756)
| table -time, Member_Account_Name, Group_Name, Subject_Account_Name
```

What was the source IP address of nathan.brooks's first TGT request?

10.5.50.12

```
index=* EventCode=4768 Account_Name=nathan.brooks
| table -time, src_ip, Account_Name, Client_Address, Ticket_E
```

ncryption_Type

New Search

1 index=* Account_Name="nathan.brooks" krbtgt

✓ 1 event (before 8/17/26 8:27:07.000 AM) No Event Sampling ▾ Job ▾ II

Events (1) Patterns Statistics Visualization

Timeline format ▾ — Zoom Out + Zoom to Selection × Deselect

Format ▾ Show: 20 Per Page ▾ View: List ▾

< Hide Fields	≡ All Fields	i	Time	Event
SELECTED FIELDS a host 1 a source 1 a sourcetype 1 INTERESTING FIELDS a Account_Name 1 a action 1 a app 1 a body 1 a category 1 a Client_Address 1 # Client_Port 1 a ComputerName 1 a dest 1 a dest_nt_domain 1 a dest_nt_host 1 a dvc 1 a dvc_nt_host 1 a Error_Code 1 # event_id 1 a event_time 1 # EventCode 1 a eventtype 7 # EventType 1		▼	2/3/26 9:13:39.000 PM	02/03/2026 01:13:39 PM LogName=Security EventCode=4768 EventType=0 ComputerName=THM-DC.tryhatmestudios.thm SourceName=Microsoft Windows security auditing. Type=Information RecordNumber=71749 Keywords=Audit Success TaskCategory=Kerberos Authentication Service OpCode=Info Message=A Kerberos authentication ticket (TGT) was requested. Account Information: Account Name: nathan.brooks Supplied Realm Name: tryhatmestudios.thm User ID: S-1-5-21-850000725-1200463294-3662097055-2607 Service Information: Service Name: krbtgt Service ID: S-1-5-21-850000725-1200463294-3662097055-502 Network Information: Client Address: ::ffff:10.5.50.12 Client Port: 53496

Task 8 - Conclusion

- Kerberos, LDAP, SMB and RDP generate most AD traffic, and understanding them helps us interpret events.
- Authentication events appear on DCs, logon sessions appear on targets, and we need both.
- Without proper configuration, many events aren't logged at all.
- Thousands of events per day are expected. Use stack counting to find anomalies.
- Individual events are data, but correlated sequences reveal what actually happened.